

Validation-Swap Distributions

Diagnosing Machine-Scored Outcomes in Experiments

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Abstract

In an experiment with clustered treatment assignment, a machine-learning or large language model scores every outcome, and human labels are affordable for only some clusters. Before coding begins, the researcher must decide how many clusters to validate, and the decision requires knowing what each budget buys. Two-phase and prediction-powered estimators correct the treatment effect with the labeled clusters, and power analysis prices a labeling design by the estimator’s asymptotic variance. When scoring error is concentrated in a few clusters, the variance misprices the risk: the corrected estimate’s distribution across possible label sets is discrete and far from Normal, with most draws moving the estimate not at all and a few moving it far. We price the design by that distribution itself. For each budget k we swap human labels into a set of k clusters, recompute the treatment effect, and enumerate all sets exactly, which is feasible in benchmarks, pilots, and fully coded studies used to plan smaller budgets. As k grows, the mean of the distribution moves on a straight line from the machine-scored estimate to the fully human-coded one, whatever the error structure, so everything that distinguishes labeling designs lies in the dispersion and tails. In a worked example, two scoring systems agree on RMSE, R^2 , both endpoints, and the entire mean path, yet when the error sits in two of twenty clusters, labeling five clusters leaves the estimate unchanged 55.3% of the time. With a single real validation draw the full distribution is out of reach, but its variance at every budget remains estimable without bias, and a negligibility condition says when a Normal approximation can be trusted: only when no small set of clusters dominates the correction. The distribution conditions on the observed pool of gold labels; inference about the population still requires a probability validation design or a transport model.

Researchers can score every response in an experiment with a classifier, a computer-vision model, or a large language model while paying human coders to score only a fraction (Gilardi et al., 2023). The cheap score may have excellent test-set accuracy and still distort the treatment effect. The relevant error is not average scoring error. It is the difference in scoring error between treatment arms.

Suppose an experiment tests whether a message changes the hostility expressed in an open-ended response. A language model scores hostility for all respondents, and human coders score a random subset. The model can overstate hostility by two hundredths in one arm and understate it by two hundredths in the other. Its mean error is zero, but the estimated treatment effect is wrong by four hundredths. Aggregate accuracy conceals this error because accuracy and the treatment contrast weight observations differently.

Existing methods already correct downstream inference with a small set of gold labels. Post-prediction inference, design-based supervised learning, and prediction-powered inference combine

predictions on many observations with gold-label residuals on a smaller probability sample (Wang et al., 2020; Egami et al., 2023; Angelopoulos et al., 2023a,b). Two-phase designs for allocating expensive measurements go back to Neyman (1938); modern treatments choose the validation sample to minimize the variance of a target estimator (Amorim et al., 2021), including adaptively (Zrnic and Candès, 2024). Parallel corrections exist when the machine prediction enters the analysis as a regression covariate rather than the outcome (Fong and Tyler, 2021). Work on text as data explains why a learned outcome requires separate measurement and causal design choices (Egami et al., 2022). These methods establish the estimator and its uncertainty.

This paper asks a different question: how much does the answer depend on which clusters receive the gold labels? We define the fixed-budget random-swap distribution of the treatment-effect estimate. Its mean is often uninformative. Its spread shows whether a validation budget produces a stable correction or a lottery. Its tails show how often the estimate moves by more than a prespecified substantive amount.

Throughout, treatment is assigned at the cluster level, as in a site-, classroom-, or wave-randomized experiment. That scope makes the cluster the natural unit for this question for the same reason it is the unit of inference: the treatment contrast is a difference of arm means, so scoring error distorts it only through within-cluster sums. Scoring failures also concentrate by cluster in practice: a model that misreads one site’s context is wrong for that whole site, and human recoding is organized in the same batches. A validation draw therefore reveals cluster-level contributions, and the corrected estimate inherits the randomness of which clusters were drawn. Validating individuals is the special case of clusters of size one. With many independent errors the correction is a sum of small terms, stable across draws and close to Normal, and the diagnostic adds little. The fragility this paper studies is a few-effective-units phenomenon, and cluster structure is how experiments end up with few effective units. The same construction applies to documents, sites, waves, or any other sampling unit, provided the swap unit matches the validation design and contains the error dependence.

The contribution is deliberately narrow. The design-scaled estimator below is a special case of established difference and prediction-powered estimators. The finite-population variance formula is standard sampling theory. We use these results to define an estimand-specific validation diagnostic, and we state four formal properties: a sharp bound on how little accuracy metrics constrain the treatment contrast, the exact sense in which the mean swap path is uninformative, design-unbiased estimation of the full budget-variance path from one validation draw, and the negligibility condition under which moment summaries with a Normal approximation suffice. Each is a short application of standard results; together they show why an average swap path can hide the feature an audit is meant to find.

1 Machine scores as experimental outcomes

Index experimental clusters by $g = 1, \dots, G$ and observations within clusters by i . Treatment $D_g \in \{0, 1\}$ is assigned at the cluster level. Let Y_{gi} be the human-coded outcome and $\hat{Y}_{gi}$ the machine score. Define the observed scoring discrepancy

$$e_{gi} = Y_{gi} - \hat{Y}_{gi}. \tag{1}$$

The machine score must be fixed independently of the outcome analysis. A score learned on the experimental outcomes requires a separate training sample or an appropriate cross-fitting design. Swaps do not repair overfitting.

For clarity, consider the unadjusted difference in means. Write N_1 and N_0 for the numbers of observations in treated and control clusters. The proxy and gold estimates are

$$\hat{\tau}_P = \frac{1}{N_1} \sum_{g:D_g=1} \sum_i \hat{Y}_{gi} - \frac{1}{N_0} \sum_{g:D_g=0} \sum_i \hat{Y}_{gi}, \quad (2)$$

$$\hat{\tau}_G = \frac{1}{N_1} \sum_{g:D_g=1} \sum_i Y_{gi} - \frac{1}{N_0} \sum_{g:D_g=0} \sum_i Y_{gi}. \quad (3)$$

Random treatment assignment gives these contrasts their experimental interpretation. Validation sampling is a second design layered on top of the experiment. It determines what can be learned about $\hat{\tau}_G - \hat{\tau}_P$ when Y is expensive.

Define cluster g 's contribution to this endpoint gap as

$$\psi_g = \begin{cases} N_1^{-1} \sum_i e_{gi}, & D_g = 1, \\ -N_0^{-1} \sum_i e_{gi}, & D_g = 0. \end{cases} \quad (4)$$

The contributions add exactly:

$$\sum_{g=1}^G \psi_g = \hat{\tau}_G - \hat{\tau}_P. \quad (5)$$

Equation (5) is an accounting identity within the paired gold-label pool. It does not identify the errors of unlabeled clusters. For OLS with fixed covariates, residualized-treatment weights replace the treated and control weights in equation (4). The same adding-up and sampling identities follow. A nonlinear estimator can still be recomputed over swap sets, but its mean path need not be linear.

The introduction's claim that aggregate accuracy cannot certify the treatment contrast can be made exact. Let $N = N_1 + N_0$ and let $r = (N^{-1} \sum_g \sum_i e_{gi}^2)^{1/2}$ be the root mean squared error of the machine score on the experimental sample.

Proposition 1 (Sharp accuracy bound). *For every error configuration with root mean squared error r ,*

$$|\hat{\tau}_G - \hat{\tau}_P| \leq r \sqrt{N \left(\frac{1}{N_1} + \frac{1}{N_0} \right)}, \quad (6)$$

with equality exactly when e_{gi} is proportional to $1/N_1$ on treated observations and $-1/N_0$ on control observations.

Proof. The gap $\hat{\tau}_G - \hat{\tau}_P = \sum_g \sum_i w_{gi} e_{gi}$ is linear in the errors, with weights $w_{gi} = 1/N_1$ when $D_g = 1$ and $w_{gi} = -1/N_0$ when $D_g = 0$. By the Cauchy-Schwarz inequality its absolute value is at most $\|w\| \|e\|$, where $\|w\|^2 = 1/N_1 + 1/N_0$ and $\|e\|^2 = Nr^2$, with equality exactly when e is proportional to w . $\square$

The bound is attainable, and it is weak at the scale that matters. In the numerical design below, RMSE 0.200 permits an endpoint gap as large as 0.400, 4 times the gold treatment effect of 0.100.

Accuracy metrics constrain the contrast only at the order of r itself, while the errors that matter are the arm-aligned ones that equation (6) shows accuracy tolerates.

2 The fixed-budget swap distribution

Let S_k be a simple random sample without replacement of exactly k clusters. The raw-swap estimate replaces machine scores by human outcomes in the selected clusters:

$$\hat{\tau}_R(S_k) = \hat{\tau}_P + \sum_{g \in S_k} \psi_g. \quad (7)$$

The distribution of $\hat{\tau}_R(S_k)$ over all $\binom{G}{k}$ sets is the raw fixed-budget swap distribution. It describes partial repair of the observed data. A fixed k is preferable to independent Bernoulli swaps when k represents a coding budget.

Let $\bar{\psi} = G^{-1} \sum_g \psi_g$ and $S_\psi^2 = (G-1)^{-1} \sum_g (\psi_g - \bar{\psi})^2$. Standard sampling identities give

$$\mathbb{E}_{S_k}[\hat{\tau}_R(S_k)] = \hat{\tau}_P + \frac{k}{G}(\hat{\tau}_G - \hat{\tau}_P), \quad (8)$$

$$\mathbb{V}_{S_k}[\hat{\tau}_R(S_k)] = k \left(1 - \frac{k}{G}\right) S_\psi^2. \quad (9)$$

Equation (8) is the central result. In a linear outcome analysis, the average raw-swap path must be a straight line connecting the proxy and gold estimates. Any two scoring systems with the same endpoints have the same mean path. The path says nothing about whether the correction is spread evenly across clusters or concentrated in a few. Equation (9) shows where that information resides. Two further facts follow directly from equations (5) and (8) and need no proof beyond arithmetic. The $k=1$ distribution is uniform on $\{\hat{\tau}_P + \psi_g\}$, so it identifies the multiset of cluster contributions, and every fixed- k distribution is a symmetric function of that multiset. The mean path, by contrast, depends on the contributions only through their sum: any two error patterns with the same endpoints share the entire mean path, however differently the correction is distributed across clusters.

The distribution supports several descriptive summaries. A researcher can report its central quantiles, the chance that swapping k clusters changes the estimate's sign, and the chance that the movement exceeds a prespecified substantive threshold. These probabilities refer to the stated swap design. They are not p -values.

3 From partial swaps to a validation estimator

Raw replacement mechanically shrinks as k/G falls. If the goal is to estimate the fully human-coded contrast from a k -cluster probability sample, scale the observed correction by its inclusion probability:

$$\hat{\tau}_H(S_k) = \hat{\tau}_P + \frac{G}{k} \sum_{g \in S_k} \psi_g. \quad (10)$$

This is a Horvitz-Thompson difference correction (Horvitz and Thompson, 1952; Särndal et al., 1992). In this simple setting it is also the rectifier used by prediction-powered inference (Angelopoulos et al., 2023a). It satisfies

$$\mathbb{E}_{S_k}[\hat{\tau}_H(S_k)] = \hat{\tau}_G, \quad (11)$$

$$\mathbb{V}_{S_k}[\hat{\tau}_H(S_k)] = \left(\frac{G}{k}\right)^2 k \left(1 - \frac{k}{G}\right) S_\psi^2. \quad (12)$$

Thus the design-scaled swap distribution is centered on the gold endpoint. Its spread measures validation-sample fragility at budget k .

Unequal-probability validation replaces G/k with inverse inclusion probabilities. Stratifying the validation draw by treatment arm or by an observable proxy-risk score can improve precision. Those choices change the swap distribution and must be included in its definition. The optimal choice depends on the target estimator (Amorim et al., 2021; Zrnic and Candès, 2024).

An exact distribution requires ψ_g for every cluster in the finite pool. That is available in a benchmark dataset, a simulation, or a large pilot used to study smaller future budgets. With only one small validation sample, the analyst cannot recover the unseen cluster contributions nonparametrically. Standard design-based variance estimation remains available when selection probabilities are known. A cluster bootstrap or a model for e_{gi} can approximate more of the distribution, but then the result depends on resampling or transport assumptions.

The second moment of the swap distribution is nevertheless estimable from a single draw, at the observed budget and at every other one.

Proposition 2 (The variance path is estimable from one validation sample). *Let S_k be a simple random sample of $k \geq 2$ clusters, and let $s_\psi^2 = (k-1)^{-1} \sum_{g \in S_k} (\psi_g - \bar{\psi}_{S_k})^2$ be the sample variance of the observed contributions. Then $\mathbb{E}_{S_k}[s_\psi^2] = S_\psi^2$. Substituting s_ψ^2 into equations (9) and (12) therefore gives unbiased estimates of the raw and design-scaled swap variances at every budget $k' \in \{1, \dots, G\}$, and Chebyshev’s inequality bounds the tail probability $\Pr\{|\hat{\tau}_H(S_{k'}) - \hat{\tau}_G| > \delta\}$ by $(G/k')^2 k'(1 - k'/G) S_\psi^2 / \delta^2$ for every threshold $\delta > 0$.*

Proof. Design-unbiasedness of the sample variance under simple random sampling without replacement is classical (Cochran, 1977; Li and Ding, 2017). The variance claims substitute this estimate into the exact formulas, and the tail bound applies Chebyshev’s inequality to the design-scaled estimator, which is centered on $\hat{\tau}_G$ by equation (11). $\square$

A pilot at one budget therefore prices fragility at every other budget. This is the finite-population analogue of power analysis for prediction-powered inference, which chooses the labeled sample size from an asymptotic variance (Chen et al., 2026). The estimate is honest but noisy: the sampling variance of s_ψ^2 depends on fourth moments of the contributions, so with few validated clusters and concentrated error the estimated variance path is itself fragile.

Moment summaries also carry an implicit distributional assumption. The following negligibility condition, due to Hájek (1960) and stated here in the form of Theorem 1 of Li and Ding (2017), says when the swap distribution can be treated as Normal.

Proposition 3 (Normal approximations require diffuse contributions). *Embed the design in a sequence of finite populations with $G \rightarrow \infty$ clusters and budgets $k = k(G)$. If*

$$\frac{1}{\min(k, G - k)} \cdot \frac{\max_g (\psi_g - \bar{\psi})^2}{S_\psi^2} \rightarrow 0, \quad (13)$$

then the standardized raw-swap estimate $\{\hat{\tau}_R(S_k) - \mathbb{E}[\hat{\tau}_R(S_k)]\} / \mathbb{V}[\hat{\tau}_R(S_k)]^{1/2}$ converges in distribution to a standard Normal, and the design-scaled estimate, an affine transformation at fixed k , converges with it. The condition fails when a bounded number of clusters carries a nonvanishing share of the centered squared contributions.

Proof. The raw-swap estimate is $\hat{\tau}_P$ plus k times the sample mean of a simple random sample of the finite population $\{\psi_1, \dots, \psi_G\}$, so the claim is the finite population central limit theorem for simple random sampling (Erdős and Rényi, 1959; Hájek, 1960), in the sufficient form of Theorem 1 of Li and Ding (2017). Sharp necessary and sufficient conditions of Lindeberg type are in Hájek (1960). $\square$

The finite-sample diagnostic implied by condition (13) is the largest single-cluster share of the centered squared contributions. In the numerical design below, one cluster carries 45% of that total in the concentrated case and 25% in the offsetting case, and the diffuse case is degenerate. Budget planning that combines Proposition 2 with a Normal approximation is therefore reliable exactly when the correction is diffuse, and misleading in the concentrated case the diagnostic exists to detect. That is the formal reason to examine the exact distribution rather than its first two moments.

3.1 Related subset diagnostics

Two neighboring diagnostics also perturb subsets of a completed analysis, and the contrasts locate the swap distribution precisely. A wild cluster bootstrap applies mean-zero random multipliers to cluster score contributions to approximate a null sampling distribution (Cameron et al., 2008). A validation swap applies zero-one inclusion indicators to observed gold-minus-proxy discrepancies. Its immediate target is the effect of a labeling design, not repeated-sample uncertainty under a null.

Dropping-data robustness metrics ask whether removing a small, adversarially chosen set of observations changes a conclusion, and locate that set with a first-order influence approximation because exact enumeration is infeasible (Giordano et al., 2026). The swap construction differs on all three choices: the perturbed unit is the gold-label set rather than the data, the perturbation is a design-random draw rather than an adversarial deletion, and no approximation is needed because the linear contrast makes every swap effect exactly additive in the cluster contributions. The adversarial variant is nevertheless free here: the most adverse k -cluster validation draw is the sum of the k most extreme contributions, read off by sorting the ψ_g .

For the linear contrast,

$$\hat{\tau}_R(S_k) - \mathbb{E}_{S_k}[\hat{\tau}_R(S_k)] = \sum_{g=1}^G \left(Z_g - \frac{k}{G} \right) \psi_g,$$

where $Z_g = \mathbf{1}\{g \in S_k\}$. This centered expression has the form of a finite-population multiplier

process. The resemblance does not turn swap quantiles into confidence intervals. Valid uncertainty for the experimental effect must account for treatment assignment, and valid uncertainty for the correction must account for the actual validation design.

4 Same accuracy and endpoints, different swap distributions

The numerical example has 20 equal-size clusters with 50 observations each. Half the clusters are treated. The fully human-coded treatment effect is 0.100, and the machine-scored effect is 0.060 in the diffuse and concentrated cases. All three scoring systems have RMSE 0.200 and the same R^2 because they score the same gold outcomes and have the same squared-error total.

The error patterns differ only in their cluster contributions. In the diffuse case, each cluster contributes 0.040/20 to the endpoint gap. In the concentrated case, two clusters contribute 0.040/2 each and the others contribute zero. The offsetting case has two positive and two negative contributions of equal size, so the proxy and gold endpoints agree even though individual swaps can move the estimate.

Table 1: Matched prediction accuracy with different contribution patterns

Error pattern	Gold ATE	Proxy ATE	Gap	RMSE	R^2	Top-two share
Diffuse	0.100	0.060	0.040	0.200	0.965	10%
Concentrated	0.100	0.060	0.040	0.200	0.965	100%
Offsetting	0.100	0.100	0.000	0.200	0.965	50%

Note: Gold ATE and proxy ATE are differences in means. Gap is gold minus proxy. Top-two share is the two largest absolute cluster contributions divided by the sum of absolute contributions. Each row uses the same gold outcomes. Quantities are deterministic for the generated finite population.

The first two patterns share the entire mean swap path, so no average over validation draws can separate them. Every difference between them is distributional. Table 2 prices that difference.

5 What a validation budget can deliver

Table 2 prices each validation budget two ways. Tail risk is the exact share of validation sets whose scaled estimate lands more than 0.050 from the gold endpoint. The Normal price is the same probability computed from a Normal approximation with the correct finite-population variance of equation (12), the summary a variance-based power analysis would use. The two prices agree only in the diffuse case, where every draw picks up the same correction and both are zero at every budget.

Concentrated error breaks the approximation in both directions. At 5 labeled clusters the Normal price is 29% against an exact 5.3%, because 55.3% of the draws contain neither high-error cluster and leave the estimate at the machine-scored value. The exact price is also not monotone in the budget: the distribution’s few atoms cross the threshold at different budgets as the G/k scaling shrinks, so offsetting risk climbs to 62% at 7 labeled clusters before falling, while the Normal price declines smoothly and understates the risk where it is largest. A larger budget can carry more tail risk than a smaller one, and the variance-based price never shows it. Decision reversals price the same way: 10% of single-cluster draws in the offsetting case reverse the sign of the estimated

effect, the analogue of the sign-change criterion in dropping-data robustness (Giordano et al., 2026). Bias is omitted from the table because it is zero up to numerical precision for every cell. A narrow distribution does not prove that human labels measure the intended construct. It says only that the specified validation draw estimates the observed gold-label endpoint consistently.

Table 2: Exact design-scaled distributions at selected budgets

Error pattern	Gold clusters	Mean	SD	5th	95th	Tail risk	Normal price
Diffuse	2	0.100	0.000	0.100	0.100	0%	0%
	5	0.100	0.000	0.100	0.100	0%	0%
	10	0.100	0.000	0.100	0.100	0%	0%
	20	0.100	0.000	0.100	0.100	0%	0%
Concentrated	2	0.100	0.083	0.060	0.260	19%	54%
	5	0.100	0.048	0.060	0.220	5%	29%
	10	0.100	0.028	0.060	0.140	0%	7%
	20	0.100	0.000	0.100	0.100	0%	0%
Offsetting	2	0.100	0.123	-0.100	0.300	35%	68%
	5	0.100	0.071	0.020	0.180	57%	48%
	10	0.100	0.041	0.020	0.180	14%	22%
	20	0.100	0.000	0.100	0.100	0%	0%

Note: Mean, SD, and quantiles describe equation (10) over all fixed-size cluster samples. Tail risk is $\Pr(|\hat{\tau}_H - \hat{\tau}_G| > 0.050)$, computed by exact enumeration, not Monte Carlo. Normal price is the same probability under a Normal approximation with the exact finite-population variance of equation (12). Gold ATE equals 0.100 in every scenario.

6 Use and limits

A validation-swap analysis begins with the same choices required by any two-phase design. The researcher states the causal estimand, fixes the machine scoring rule, identifies the unit sampled for human coding, and records every cluster’s inclusion probability. If treatment is assigned by cluster, swapping individual labels while treating them as independent understates the relevant variation.

The primary analysis should report a valid correction estimator and interval, using prediction-powered, design-based supervised learning, or another method whose assumptions fit the design. The swap distribution is an additional diagnostic. Before seeing it, the researcher chooses validation budgets and a substantively meaningful movement threshold. The exhibit then reports quantiles, tail risk, and any decision reversal probability. Accuracy metrics remain useful descriptions of the scoring model, but they do not substitute for this coefficient-specific analysis.

The diagnostic has four limits. First, it cannot reveal the individual errors of unlabeled observations. Second, a distribution computed inside a convenience validation sample need not transport outside it. Third, human labels can contain their own error or define the wrong construct. Fourth, a small number of validated clusters may leave the distribution too poorly estimated to support fine subgroup claims. The proper conclusion in that case is that the validation budget does not resolve the scoring risk.

Machine-scored experiments create two sources of uncertainty: treatment assignment and gold-label sampling. Randomization-based or cluster-robust inference addresses the first. Established two-phase estimators address the second when they correct the machine-scored contrast. The random-

swap distribution makes validation-sample dependence visible by showing how much the conclusion depends on which clusters were checked.

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